

Total No. of Questions : 8]

SEAT No. :

PD4110

[Total No. of Pages : 2

[6402]-70

S.E. (AI&ML)

**FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

(2019 Pattern) (Semester - IV) (218553)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) What is knowledge-based agent (KBA) in AI? Explain architecture of KBA with neat diagram? [7]
b) Specify the syntax of First-order logic in BNF form with suitable example. [6]
c) Compare between Forward Chaining and Backward Chaining. [5]

OR

- Q2)** a) What is the relationship between knowledge and intelligence? Explain various levels of describing knowledge based agent. [7]
b) What is Proposition in AI? Explain types of Propositions with example. [6]
c) Discuss various approaches of designing a knowledge-based agent. [5]

- Q3)** a) Differentiate between positive class and negative class. Explain the training versus testing phase in machine learning. [4]
b) Explain different cross validation techniques. [8]
c) Discuss different applications of machine learning. [5]

OR

- Q4)** a) Elaborate various cross validation techniques with advantages and limitations. [9]
b) Explain the Machine Learning pipeline in detail with suitable diagram. [8]

- Q5)** a) Compare Supervised, Unsupervised, and Semi-Supervised learning with examples. [6]
b) What is the need for Dimensionality Reduction Explain the concept of the Curse of Dimensionality. [6]
c) State and justify Real life applications of supervised and unsupervised learning. [6]

OR

P.T.O.

- Q6)** a) Explain Principal Component analysis in brief. [6]
b) Explain Unsupervised learning with real time examples. [6]
c) Write a short note on subset selection. [6]

- Q7)** a) Define following terms: [8]
i) Accuracy
ii) Precision
iii) Recall
iv) Error rate
b) What is difference between simple linear and multiple linear regressions? Explain how to assessing performance of Regression. [9]

OR

- Q8)** a) Calculate the regression coefficient of X on Y and obtain the lines of regression for the following data [9]

X	1	2	3	4	5	6	7
Y	9	8	10	12	11	13	14

- b) What is an error w.r.t. Regression? Explain at least 4 performance measures for regression. [8]

